Lecture 12 - KNN and Clustering

ECE 447: Data Analysis and Machine Learning for Engineers

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This Lecture Covers ...

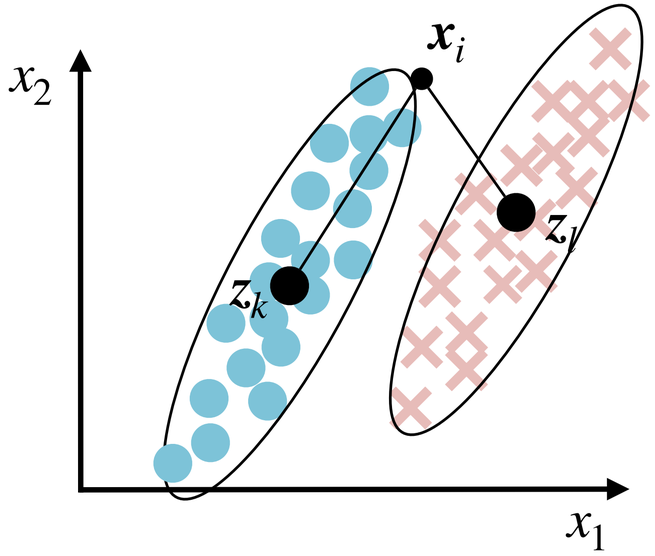
- K-Nearest Neighbour (KNN)
- Clustering Methods

Till Now, We Discussed Supervised Learning (Classification)

Minimum Euclidean Distance (MED) from prototypes of a class.

$$\boldsymbol{x}_i \in c_k \text{ iff } d(\boldsymbol{x}_i, \boldsymbol{z}_k) < d(\boldsymbol{x}_i, \boldsymbol{z}_l)$$

for all $l \neq k$

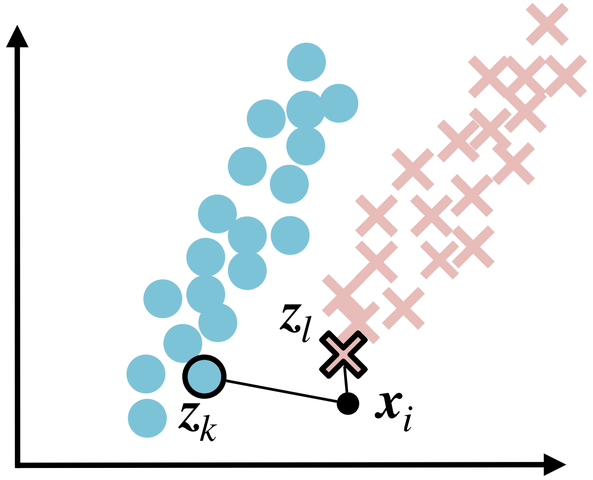


The prototype is:

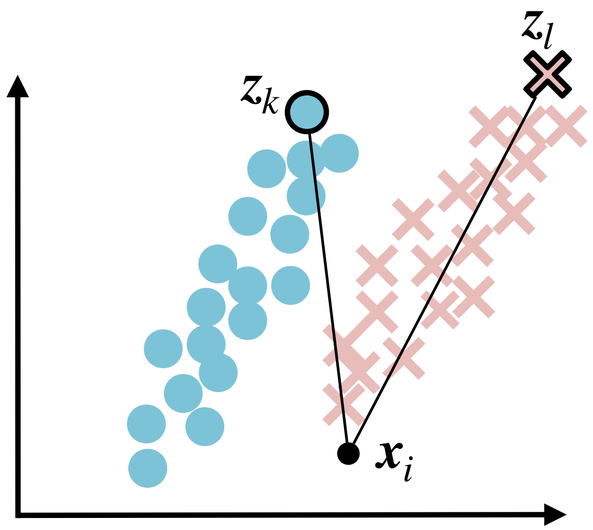
mean (centre) of each class.

Supervised Learning (Classification): Choice of the Prototype

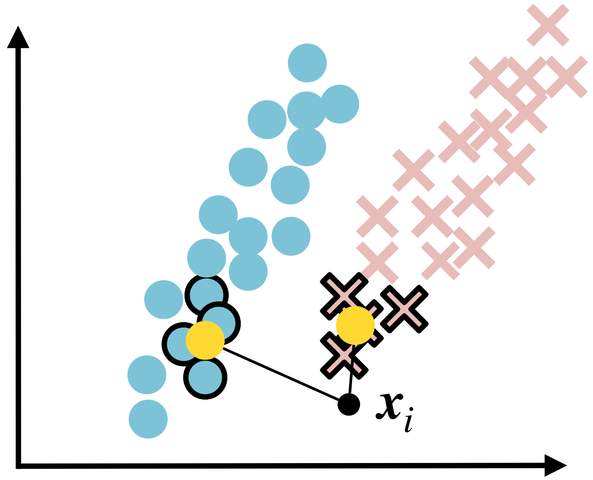
Nearest Neighbour (NN)



Farthest Neighbour

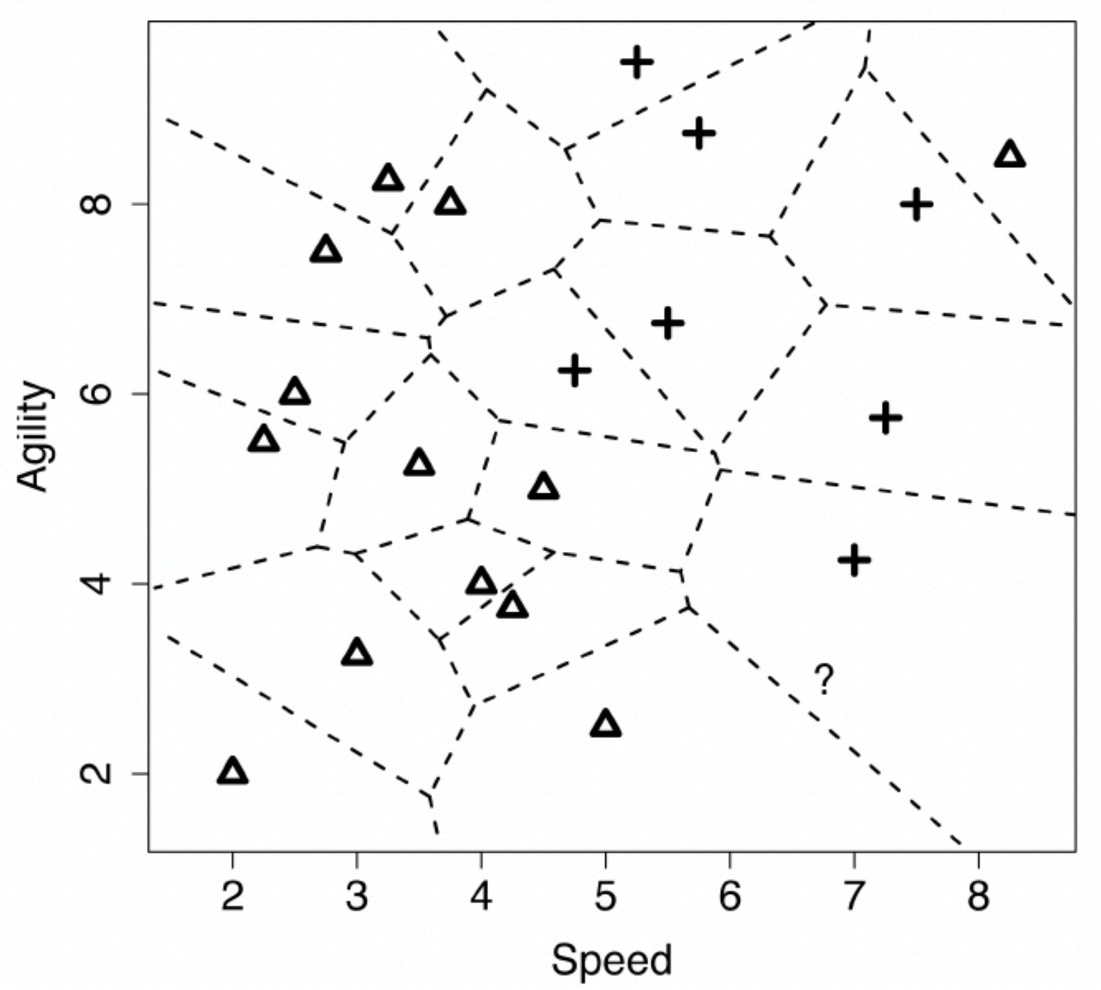


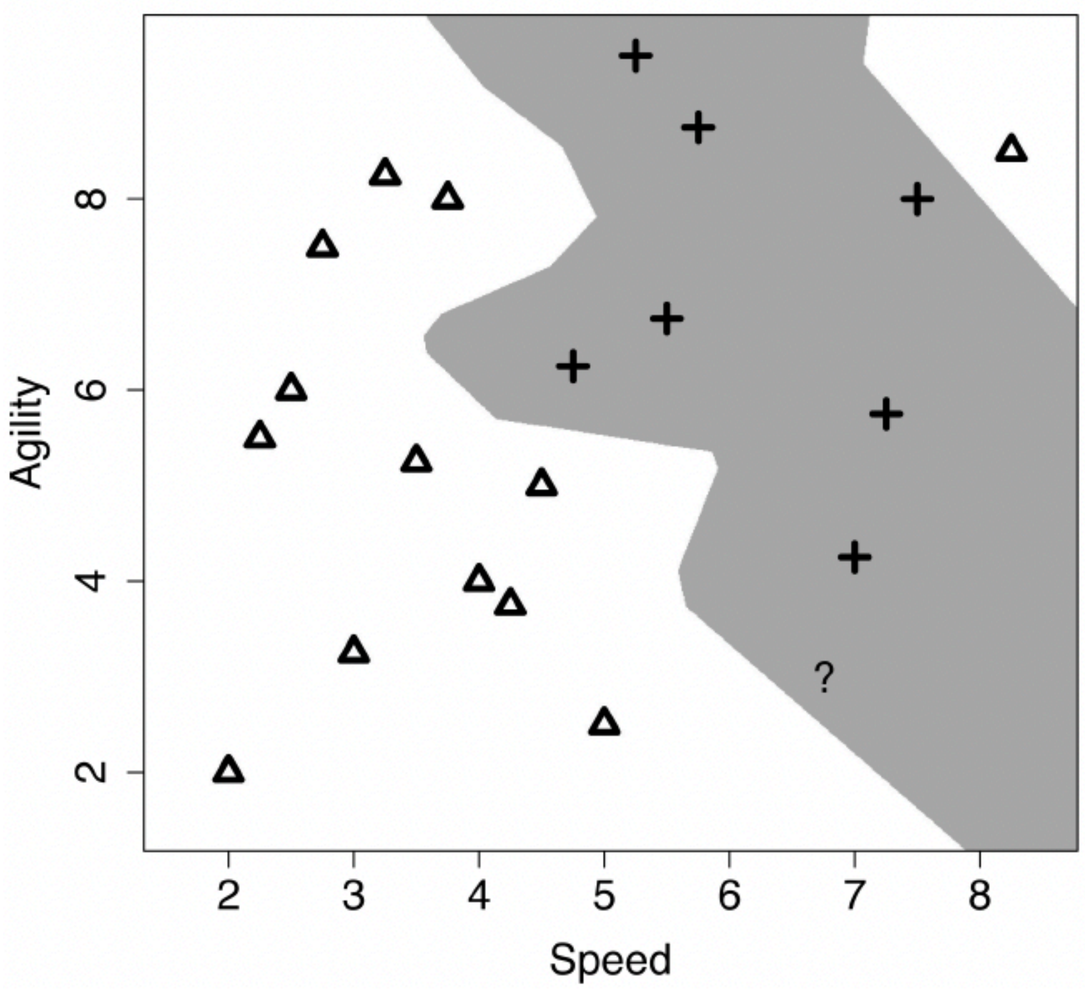
K-Nearest Neighbour (KNN)



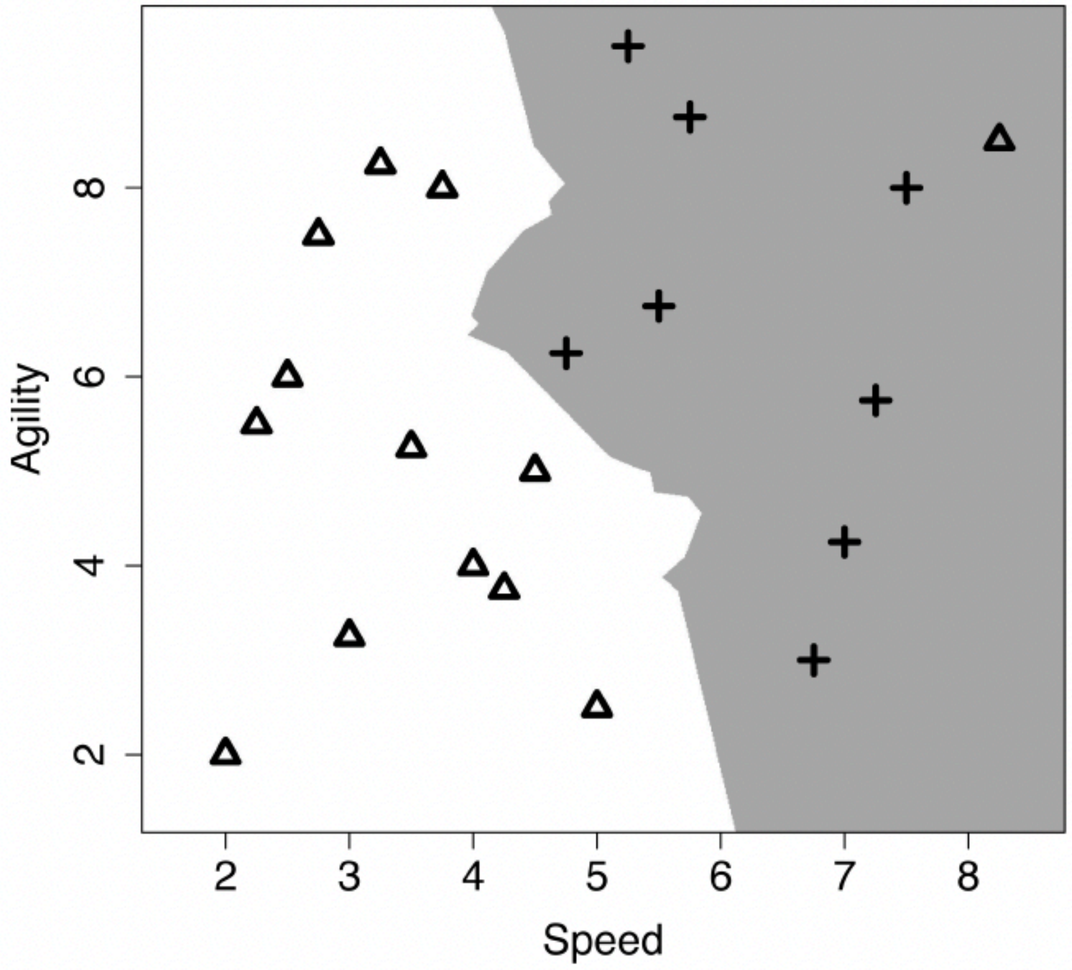
Today, we will code KNNs.

Voronoi Cells and the Decision Boundary: 2 Class, NN

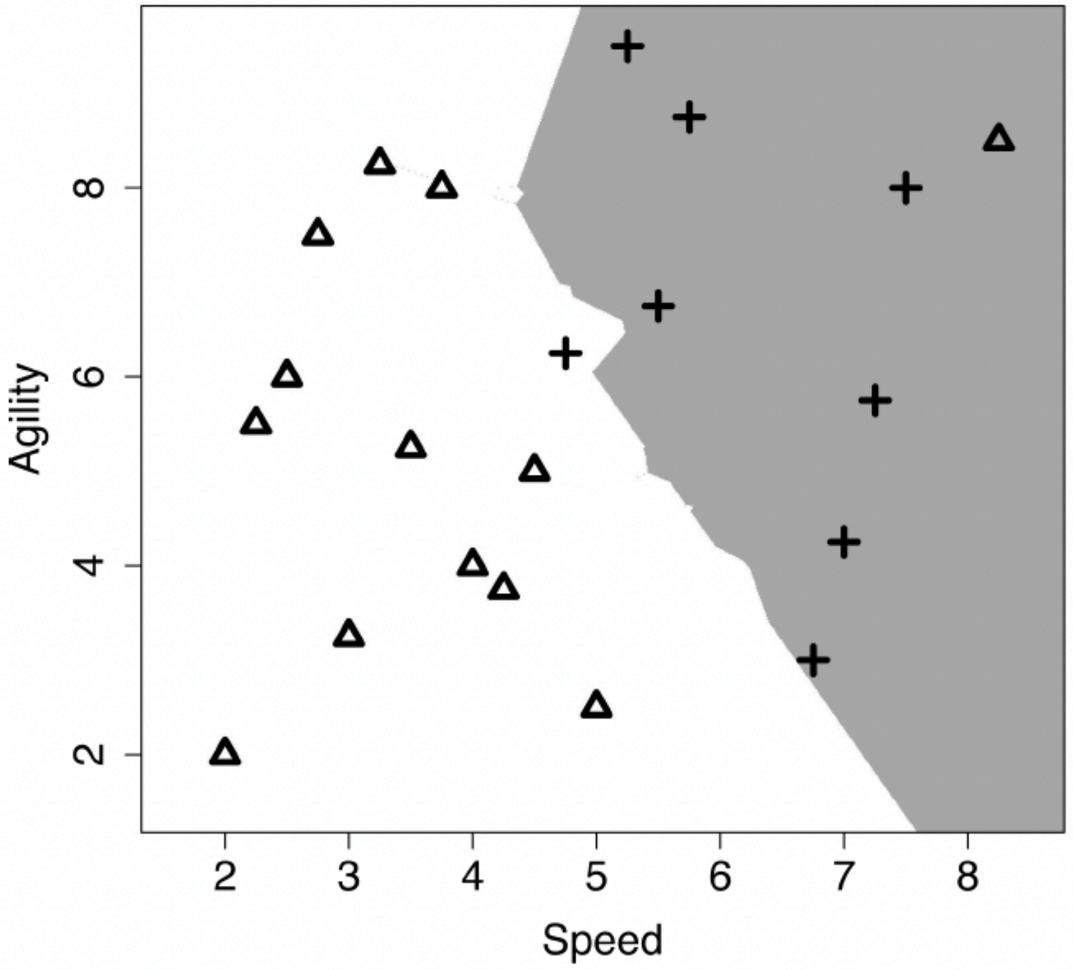




Voronoi Cells and the Decision Boundary: 2 Class, k=3,5 NN



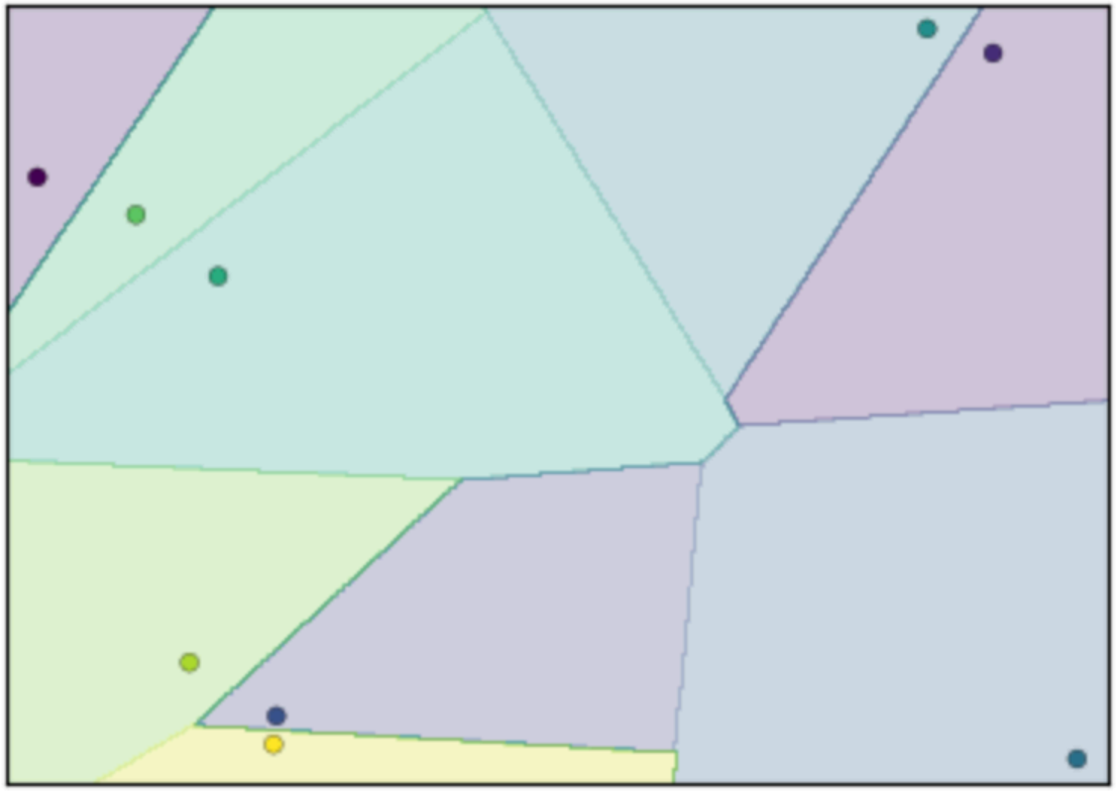
(a) Decision boundary ($k = 3$)



(b) Decision boundary ($k = 5$)

Nearest Neighbour Example with 9 Classes

Voronoi cells and decision boundary.



Unsupervised Learning with Clustering

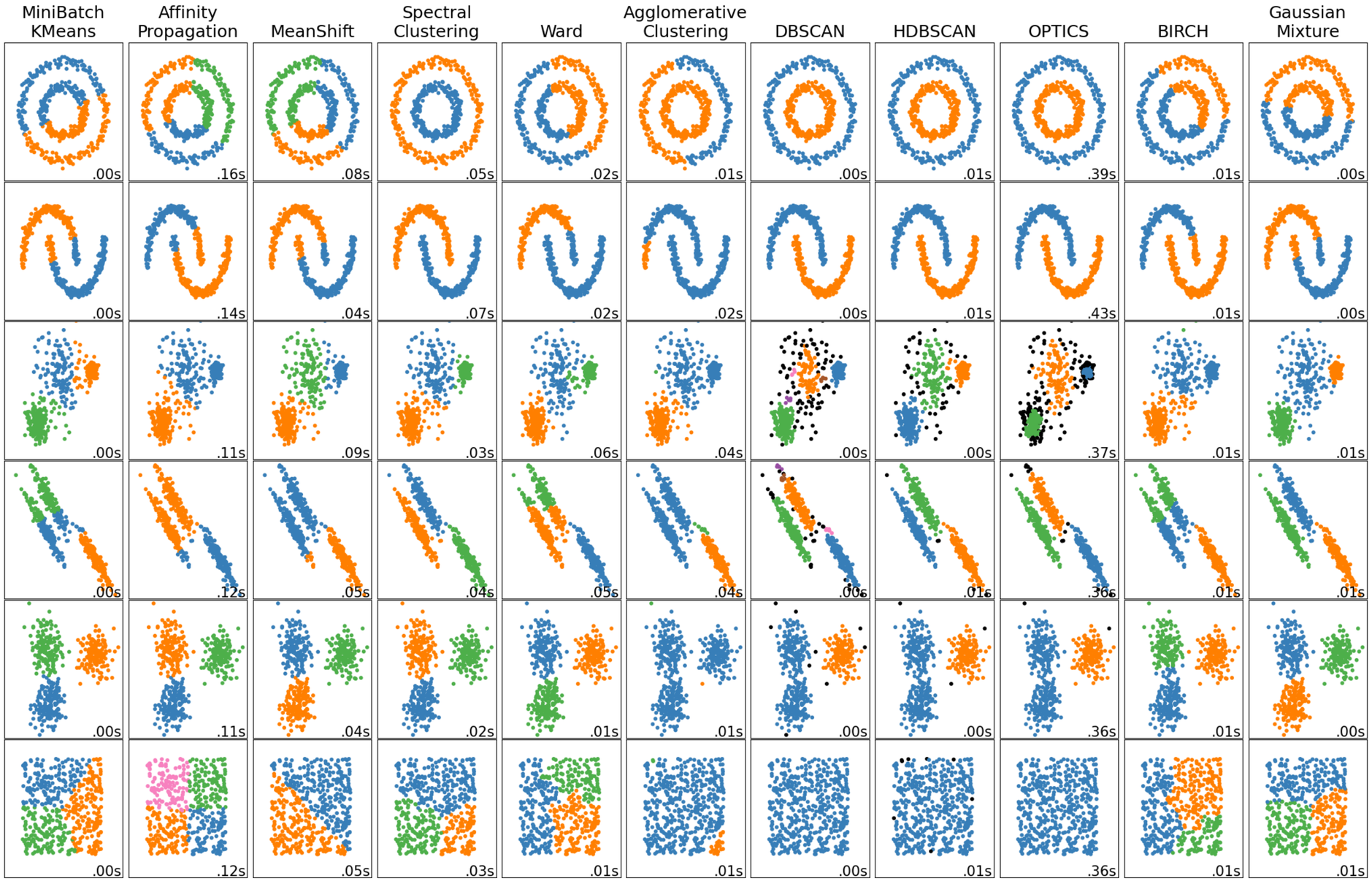
What if there is no label? We are interested in structure data into **clusters**.

Find **how many clusters** are there, and a logic to assign data to each cluster.

***Group** data example with*
similar characteristics

Overview of Clustering Methods

https://scikit-learn.org/stable/modules/clustering.html



K-Means

Clustering has an iterative (alternating) procedure.

Given an initial set of k means (a.k.a. the prototypes) $\boldsymbol{c}_1^{(1)}, \boldsymbol{c}_2^{(1)}, \ldots, \boldsymbol{c}_k^{(1)}$

1. **Assignment step**: assign each example to the clusters with the following logic:

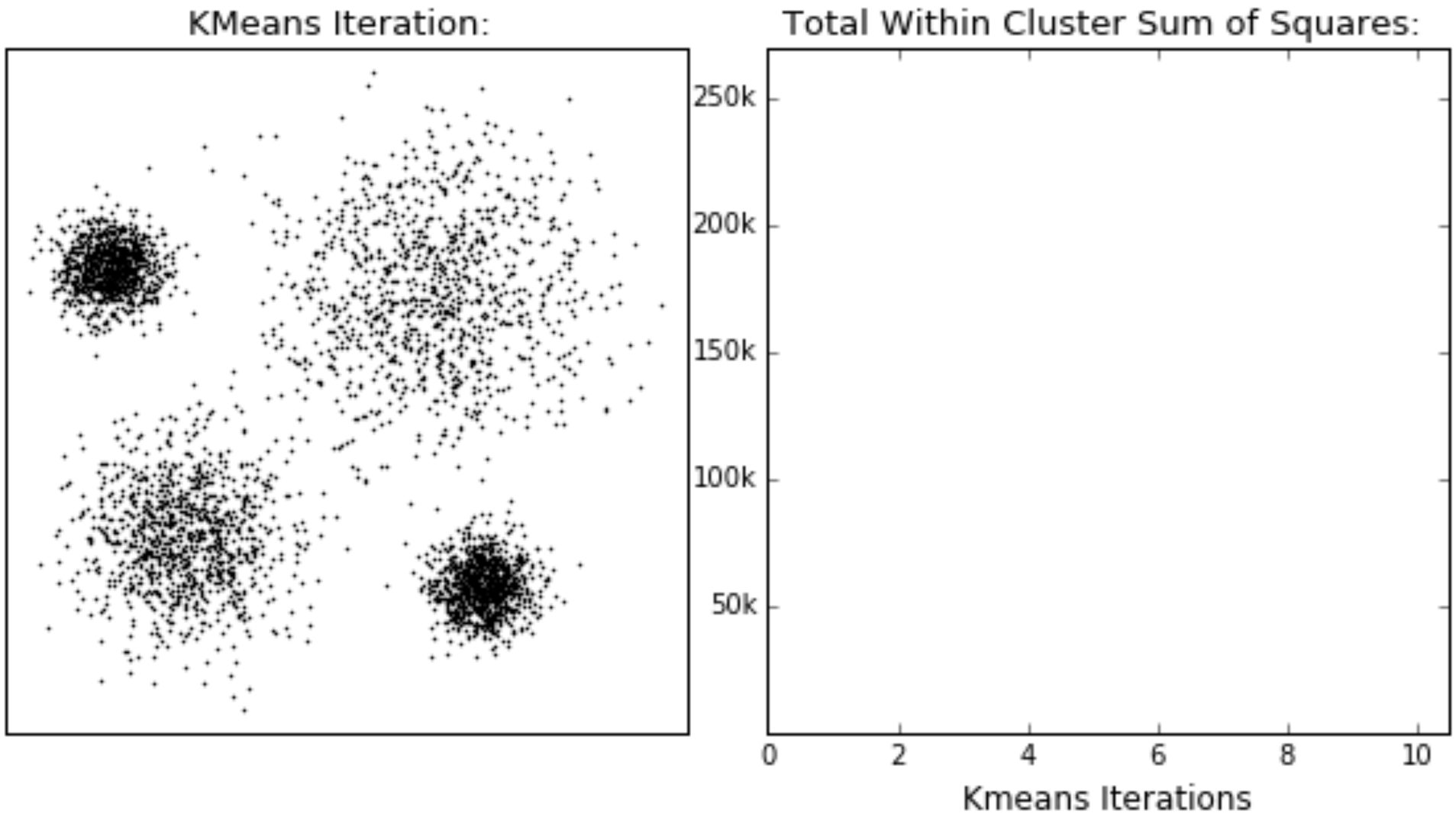
$$S_j^{(l)} = \{\boldsymbol{x}_i : \|\boldsymbol{x}_i \leq \boldsymbol{c}_j^{(l)}\|_2^2 \leq \|\boldsymbol{x}_i \leq \boldsymbol{c}_r^{(l)}\|_2^2 \quad \forall r,\ 1 \leq r \leq k\}$$

2. **Update step**: update the prototypes (centroids)

$$\boldsymbol{c}_j^{(l+1)} = \frac{1}{|S_j^{(l)}|} \sum_{\boldsymbol{x}_i \in S_j^{(l)}} \boldsymbol{x}_i$$

https://dashee87.github.io/data science/general/Clustering-with-Scikit-with-GIFs/

K-Means — Algorithm Visualization



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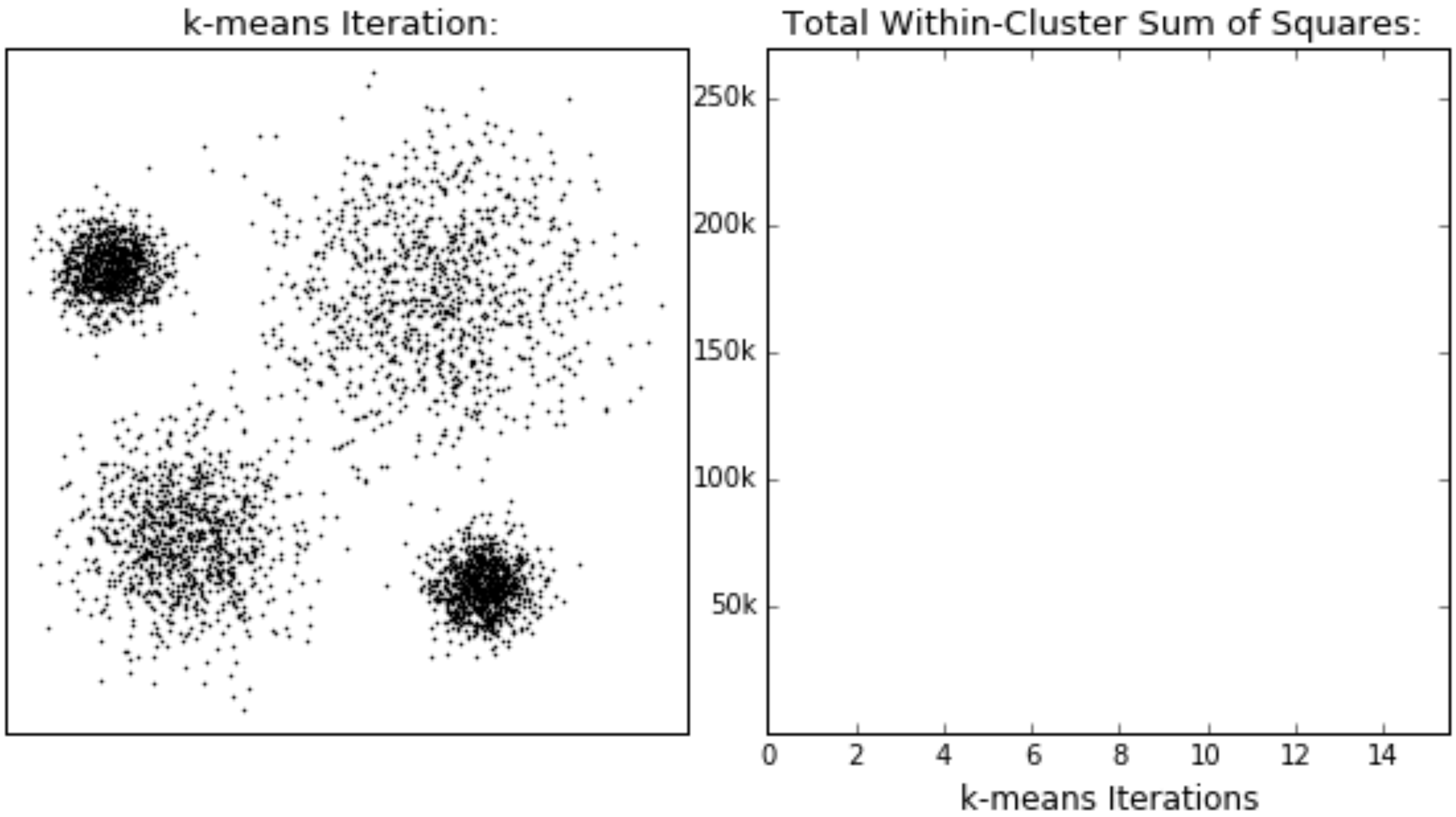
K-Means — Sensitivity to Initialization



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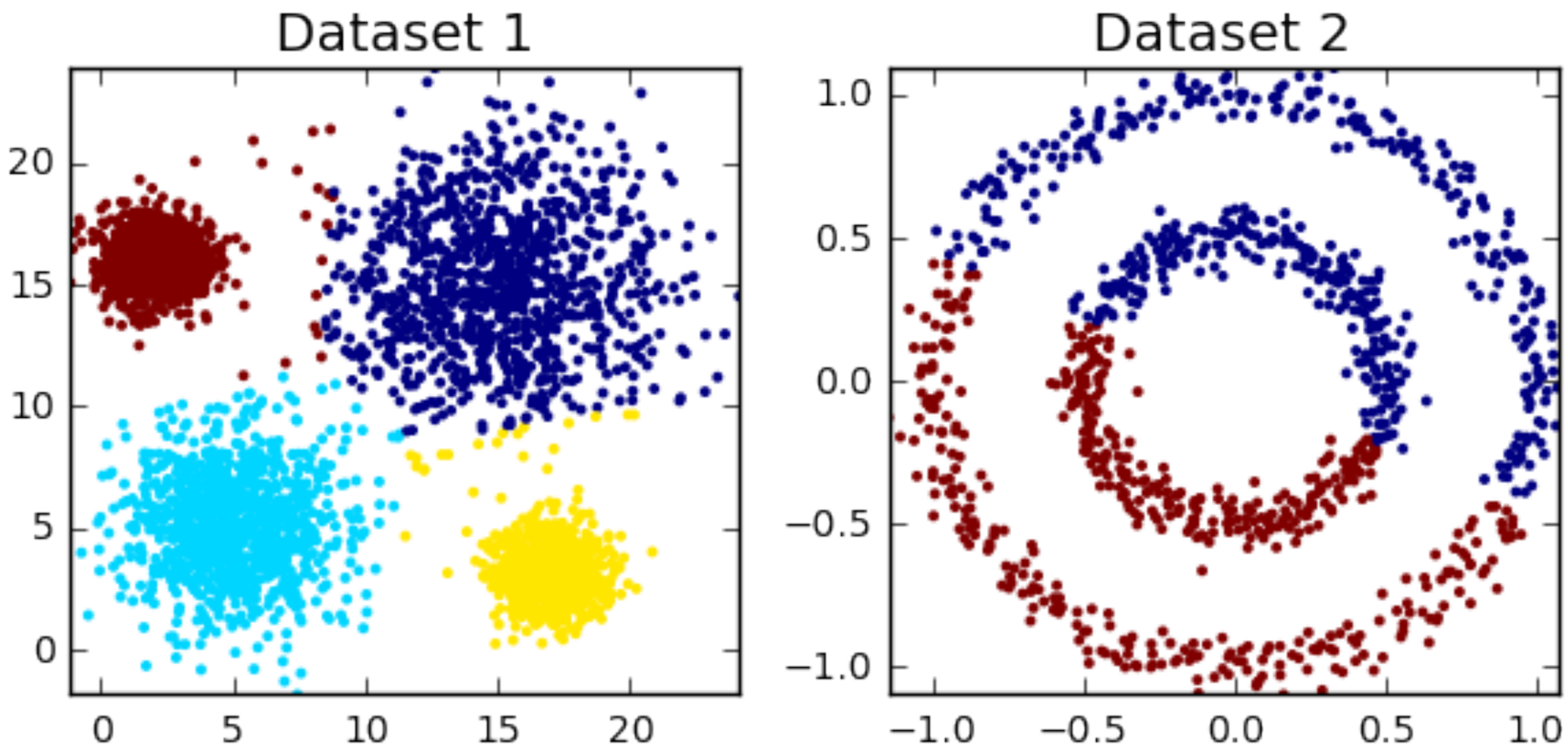
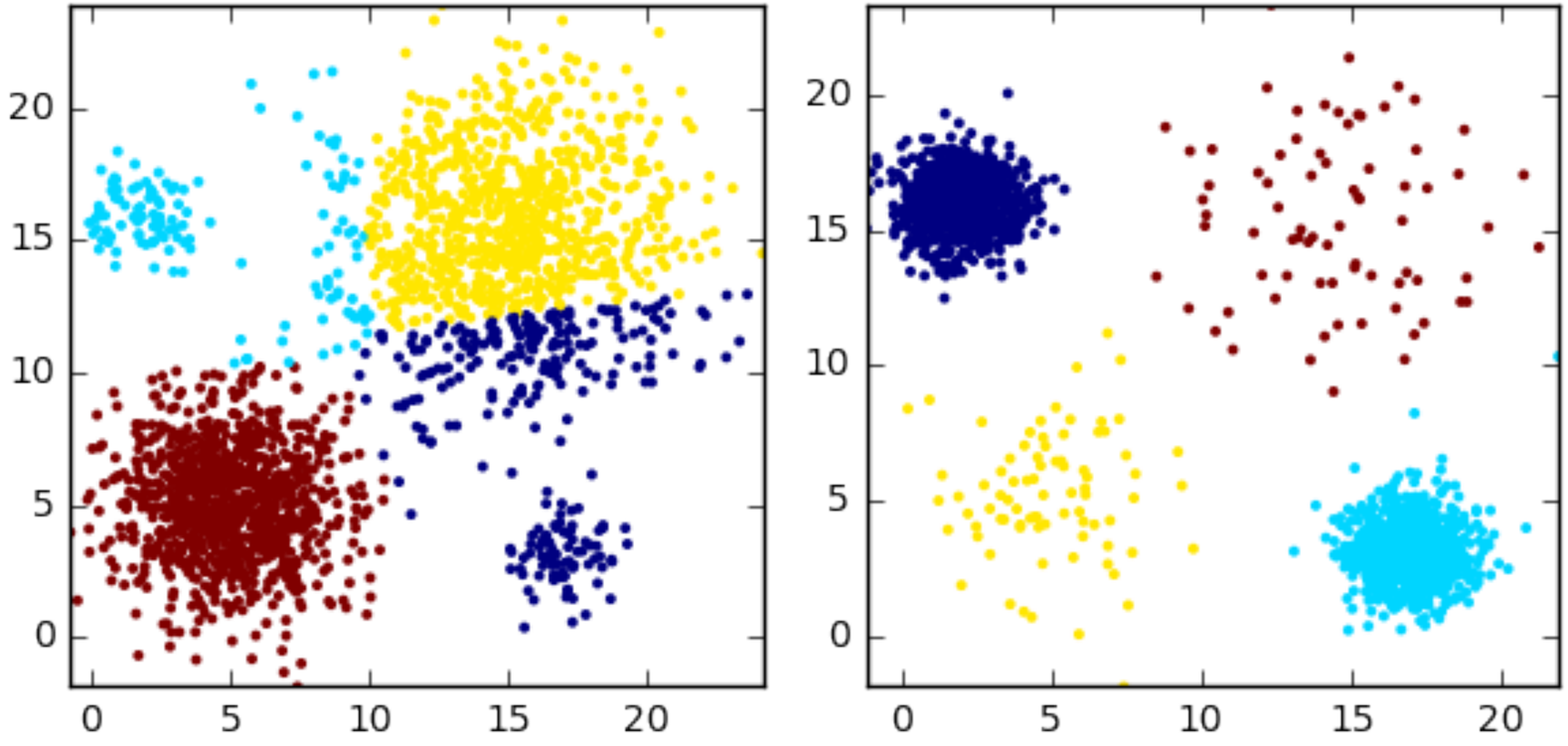
When Do K-Means Fail? (I)



*K-Means may fail when **means** are not suitable prototypes.*

https://dashee87.github.io/data science/general/Clustering-with-Scikit-with-GIFs/

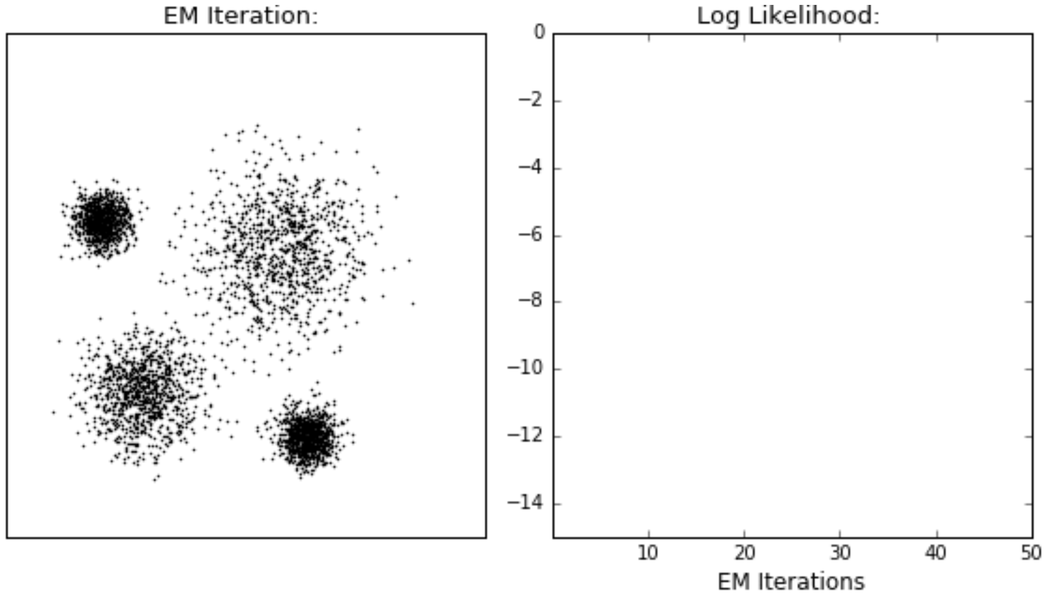
When Do K-Means Fail? (II)

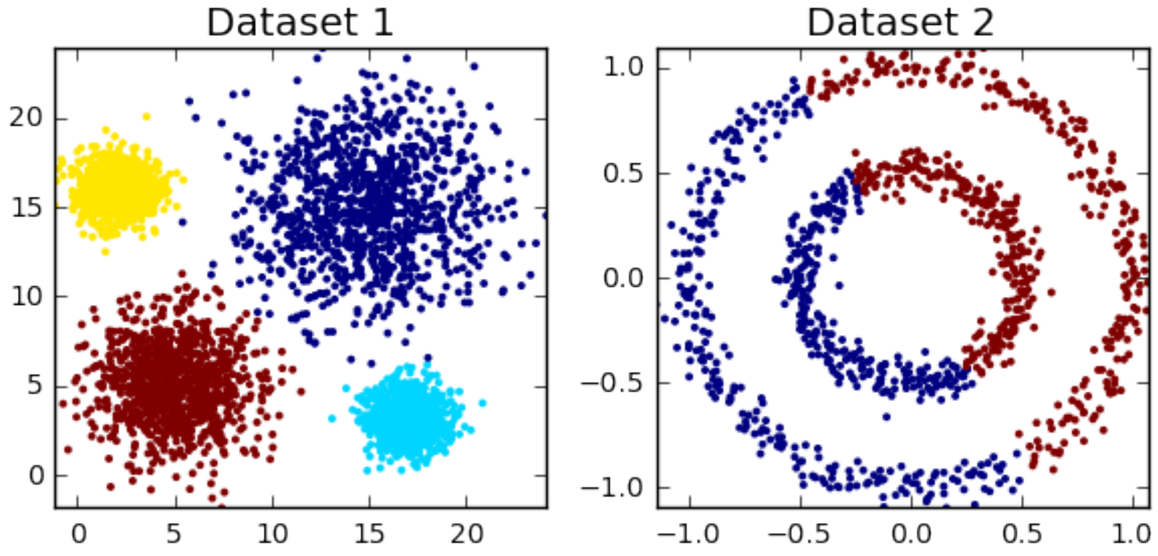


K-Means may fail when data is not balanced.

https://dashee87.github.io/data science/general/Clustering-with-Scikit-with-GIFs/

Gaussian Mixture Models (GMMs)





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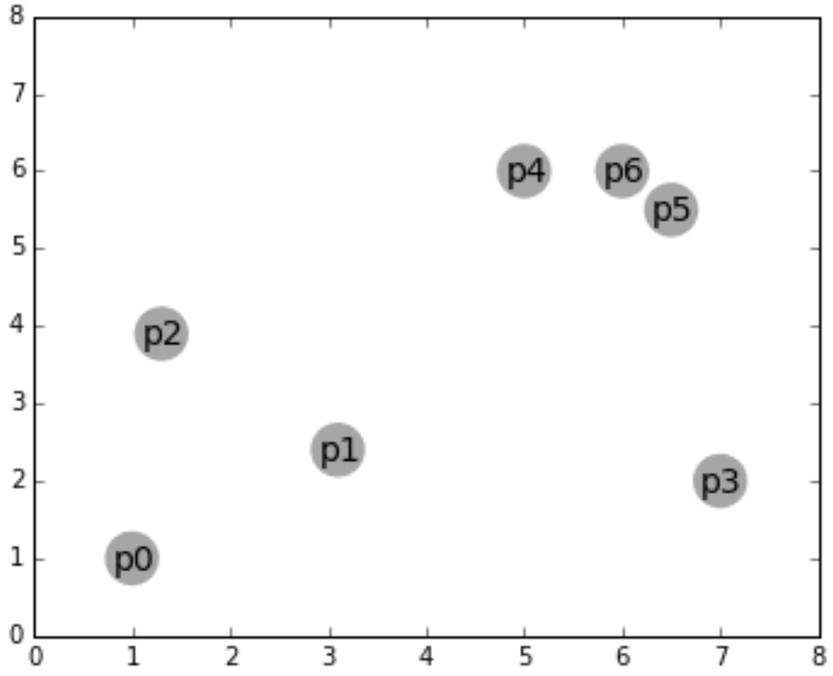
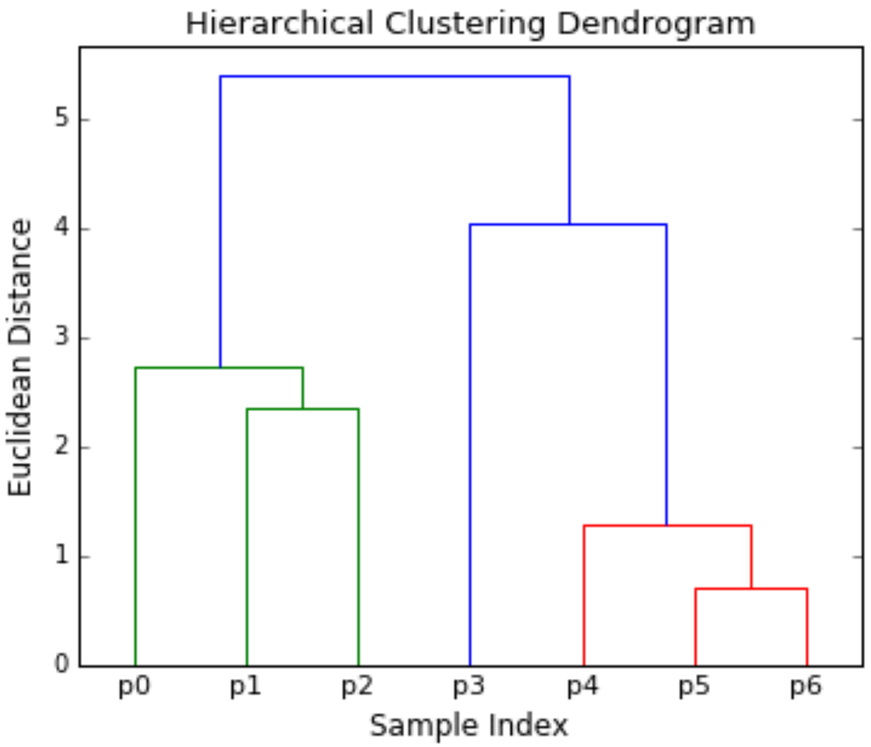
Hierarchical Clustering





Takeaways

- *KNNs split the space into Voronoi cells.*

- *Each clustering method has its own criteria (lets get intuition from the notebooks).*